

MSDM5004 Spring 2026

Exercise 3

Mar 2

Consider the equation

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}.$$

Find stability condition of the numerical scheme

$$\frac{U_j^{n+1} - U_j^{n-1}}{2\Delta t} = \frac{U_{j+1}^n - 2U_j^n + U_{j-1}^n}{(\Delta x)^2}.$$

Solution.

The numerical scheme is

$$U_j^{n+1} = U_j^{n-1} + 2\mu (U_{j+1}^n - 2U_j^n + U_{j-1}^n).$$

Let

$$U_j^n = \lambda^n e^{ik(j\Delta x)}.$$

Substitute it into the scheme,

$$\lambda^{n+1} e^{ik(j\Delta x)} = \lambda^{n-1} e^{ik(j\Delta x)} + 2\mu (\lambda^n e^{ik((j+1)\Delta x)} - 2\lambda^n e^{ik(j\Delta x)} + \lambda^n e^{ik((j-1)\Delta x)}).$$

Divide it by $\lambda^n e^{ik(j\Delta x)}$,

$$\lambda = \lambda^{-1} + 2\mu (e^{ik\Delta x} - 2 + e^{-ik\Delta x}).$$

$$\lambda = \lambda^{-1} + 2\mu (2 \cos k\Delta x - 2).$$

$$\lambda = \lambda^{-1} - 8\mu \sin^2 \frac{k\Delta x}{2}.$$

$$\lambda^2 + 8\lambda\mu \sin^2 \frac{k\Delta x}{2} - 1 = 0.$$

We have

$$\lambda_1 = -4\mu \sin^2 \frac{k\Delta x}{2} + \sqrt{16\mu^2 \sin^4 \frac{k\Delta x}{2} + 1},$$

$$\lambda_2 = -4\mu \sin^2 \frac{k\Delta x}{2} - \sqrt{16\mu^2 \sin^4 \frac{k\Delta x}{2} + 1}.$$

Since

$$|\lambda_2| = 4\mu \sin^2 \frac{k\Delta x}{2} + \sqrt{16\mu^2 \sin^4 \frac{k\Delta x}{2} + 1} > 1, \text{ when } \sin \frac{k\Delta x}{2} \neq 0,$$

This numerical scheme is always unstable.